KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4081 | 4 Credits: 4 | Program Core | Source-grounded

# Process Variables Measurement

 To impart students with the fundamental concepts of pressure measurement and its

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4081 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 4081.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

# Official course information

### Course information

| Item | Official information |
|---|---|
| Programme | Diploma in Instrumentation Engineering |
| Course code | 4081 |
| Course title | Process Variables Measurement |
| Semester | 4 Credits: 4 |
| Course category | Program Core |
| Periods per week | 4 (L:3 T:1 P:0) Periods per semester: 60 |
| Periods per semester | 60 |
| Credits | 4 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

12 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

STUDY METHOD

# How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

-  To impart students with the fundamental concepts of pressure measurement and its
- applications in industries.
-  To explain the construction, working principle and applications of different types of
- flow meters.
-  To give an insight about different techniques and applications of level measurement.
-  To illustrate the fundamental concepts and applications of different types of
- temperature measuring techniques.

### Course outcomes

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | applications of different types of pressure 15 Applying measuring instruments. Explain the construction, working principle and | See official syllabus |
| CO2 | 15 Understanding applications of different types of flow meters. Explain the different techniques for level | See official syllabus |
| CO3 | 12 Understanding measurement. Apply different types of transducers for | See official syllabus |
| CO4 | 16 Applying temperature measurement | See official syllabus |

### CO–PO mapping evidence

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 3 |
| CO2 | CO2 2 |

| Row | Extracted official mapping |
|---|---|
| CO3 | CO3 2 |
| CO4 | CO4 3 |

COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

### Module coverage

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | different types of pressure measuring instruments. | 6 | As specified |
| Module 2 | types of flow meters. | 2 | As specified |
| Module 3 | Explain the different techniques for level measurement. | 2 | As specified |
| Module 4 | Apply different types of transducers for temperature measurement | 2 | As specified |

MODULE 1

# different types of pressure measuring instruments.

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: different types of pressure measuring instruments.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## ▼ 2 Applying standards. Describe the construction and working

2 Applying standards. Describe the construction and working is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Applying standards. Describe the construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 applying standards. describe the construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
| --- | --- |

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Applying standards. Describe the construction and working means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 2 Applying standards. Describe the construction and working □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### ▼ 3 Understanding principle of different types of manometers Explain the construction and working

3 Understanding principle of different types of manometers Explain the construction and working is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding principle of different types of manometers Explain the construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding principle of different types of manometers explain the construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

##### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding principle of different types of manometers Explain the construction and working means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

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**Malayalam support:** Module 1-□□ 3 Understanding principle of different types of manometers Explain the construction and working □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

### ▼ 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for

3 Understanding principle of elastic pressure gauges Compare and apply different transducers for is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding principle of elastic pressure gauges compare and apply different transducers for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

##### Topic study guide

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

### ▼ 3 Applying various pressure measurements. Explain different methods used for vacuum

3 Applying various pressure measurements. Explain different methods used for vacuum is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Applying various pressure measurements. Explain different methods used for vacuum using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 applying various pressure measurements. explain different methods used for vacuum should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Applying various pressure measurements. Explain different methods used for vacuum means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 3 Applying various pressure measurements. Explain different methods used for vacuum □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## ▼ 2 Understanding measurement Calibrate various pressure measuring

2 Understanding measurement Calibrate various pressure measuring is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding measurement Calibrate various pressure measuring using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding measurement calibrate various pressure measuring should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
| --- | --- |

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding measurement Calibrate various pressure measuring means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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**Malayalam support:** Module 1-□□ 2 Understanding measurement Calibrate various pressure measuring □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

**▼ 2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure – simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different**

2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure – simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure – simple conversion problems- Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding instruments. definition of pressure - different types of pressure - units of pressure – simple conversion problems- methods of pressure measurement- manometers- u tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems- elastic pressure transducers-c-type and spiral bourdon gauges - bellows – diaphragms-electrical pressure transducers- strain gauges- capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- differential pressure transmitter -vacuum gauges- mcleod gauge, pirani gauge - ionization gauge -testing and calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. i/ p and p/i converters using flapper nozzle system. explain the construction, working principle and applications of different should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure – simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers-simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester-manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure – simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter - Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# types of flow meters.

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

| **As specified** | **CO-linked** |
|---|---|
| official hours | see outcomes |

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: types of flow meters.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

### ▾ 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring

4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring is an official topic in Module 2 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 4 understanding measurement. compare the construction and working principle of different types of flow 8 understanding m2.02 measurement techniques. calibrate various flow measuring should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**▼ 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli’s equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method.**

3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli’s equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. is an official topic in Module 2 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli’s equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 3 understanding instruments. principles of flow measurements- laminar and turbulent flow – reynold's number - continuity equation - bernoulli’s equation – simple problems-methods of flow measurement- variable head flow meter- orifice plate flow meter - venturi meter - flow nozzle – pitot tube – simple problems-positive displacement flow meter - nutating disc -. variable area flow meter - rotameter- coriolis type mass flow meter. electromagnetic flow meters. - ultrasonic flow meters-calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli’s equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters- Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. means in the official course context |
| Study lens | Principle → components or stages → result → application |

| Aspect | What to prepare |
|---|---|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli’s equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# Explain the different techniques for level measurement.

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Explain the different techniques for level measurement.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

**▼ principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers**

principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers is an official topic in Module 3 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, principle of different types of level 8 understanding measurement techniques. compare and apply different transducers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ **for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems.**

for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems. is an official topic in Module 3 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, for various level measurements. methods of liquid level measurement- sight glass technique - float type - displacer and torque tube type – hydro static pressure type- . air purge type level indicator – electrical methods- capacitive level indicator- radiation level detector - ultrasonic level gauge – microwave level switches – differential pressure type level transmitter – simple problems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems. [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

# Apply different types of transducers for temperature measurement

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Apply different types of transducers for temperature measurement
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

**▼ Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying**

Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying is an official topic in Module 4 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, convert different temperature scales. 3 applying explain the construction and working principle of different types of temperature 8 understanding m4.02 measurement techniques compare and apply different transducers applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**▼ 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types ( Ni, Cu, & Pt ) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types– simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com › Vendors › Electronics › A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7Idf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level**

5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types ( Ni, Cu, & Pt ) -

characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types– simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com › Vendors › Electronics › A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7Idf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level is an official topic in Module 4 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple

problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types ( Ni, Cu, & Pt ) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types– simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com › Vendors › Electronics › A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7Idf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 for various temperature measurement definition of temperature - different temperature scales and conversions – simple problems. working of bimetallic thermometer, mercury in glass thermometer, mercury in steel thermometer-construction and working of radiation pyrometer, optical pyrometer- definition of ptc and ntc – construction and working of resistance temperature detector – different types ( ni, cu, & pt ) - characteristics of rtd – simple problems - thermistor and its types- properties- thermistor characteristics- applications of thermistor-seebeck effect - peltier effect and thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- e,j,t,k,s,r - properties of different types– simple problems- thermopile - compare the characteristics of thermocouple, rtd and thermistor. text / reference: t/r book title/author a.k. sawhney, a course in mechanical measurements and instrumentation, r1 dhanpat rai r2 s.k singh , industrial instrumentation and control , tmh r3 r.k jain , mechanical and industrial measurements , khanna publishers r4 krishnaswami , industrial instrumentation , new age publications online resources: sl.no website link 1 https://www.youtube.com/watch?v=4knar4hzrjs 2 www.temperatures.com › vendors › electronics › a to d 3 https://www.youtube.com/watch?v=as5kzxkyt24 4 https://www.youtube.com/watch?v=k2ch1cukmgs 5 https://www.youtube.com/watch?v=jhcs7idf4au 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-mqu0xgh6ba 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types ( Ni, Cu, & Pt ) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- |

| Aspect | What to prepare |
|---|---|
| | E,J,T,K,S,R - properties of different types– simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com › Vendors › Electronics › A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7Idf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types ( Ni, Cu, & Pt ) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types– simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com › Vendors › Electronics › A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7Idf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

# Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

# Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

# Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

# Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

## Module 1

▶ **Define or explain 2 Applying standards. Describe the construction and working. (3 marks)**

▶ **Define or explain 3 Understanding principle of different types of manometers Explain the construction and working. (3 marks)**

▶ **Define or explain 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for. (3 marks)**

▶ **Define or explain 3 Applying various pressure measurements. Explain different methods used for vacuum. (3 marks)**

## Module 2

▶ **Define or explain 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring. (3 marks)**

▶ **Define or explain 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli's equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter -**

**Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method.. (3 marks)**

## Module 3

▶ **Define or explain principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers. (3 marks)**

▶ **Define or explain for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems.. (3 marks)**

## Module 4

▶ **Define or explain Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying. (3 marks)**

▶ **Define or explain 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types ( Ni, Cu, & Pt ) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law**

**of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types– simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com › Vendors › Electronics › A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7Idf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level. (3 marks)**

PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **2 Applying standards. Describe the construction and working**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **principle of different types of**

## Module 4

1-mark definition: State the meaning of **Convert different temperature**

**level 8 Understanding measurement techniques. Compare and apply different transducers**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

**scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

# Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

# Glossary

### Study glossary

| Term | Meaning in this lesson |
|---|---|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## Web resources listed in the official PDF


- Sl.No Website Link https://www.youtube.com/watch?v=4knAR4HzRJs
- www.temperatures.com › Vendors › Electronics › A To D https://www.youtube.com/watch?v=As5kzxkyT24 https://www.youtube.com/watch?v=K2CH1cUkMgs https://www.youtube.com/watch?v=JHCS7Idf4aU nptel.ac.in/courses/101106040/chapter 5.pdf https://www.youtube.com/watch?v=-MQU0xgh6bA
- https://instrumentationtools.com/dp-transmitters-level-measurement
- https://www.emerson.com/.../measurement-instrumentation/level


IN-PAGE SEARCH

# Search this lesson